Session 12A1: Complex Numbers — Beyond the Real Line

Phase 2 — Classical Techniques | 70 min

Part A: The Basics — i and Its Arithmetic

Example 1: i and Its Powers — A 4-Cycle

$i^2 = -1$. This single fact defines everything about complex numbers.

Multiply by i repeatedly and watch the cycle:
$i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, ...

The cycle repeats every 4 steps: $i \to -1 \to -i \to 1 \to i \to \cdots$

To find i^{50}: divide 50 by 4. Remainder 2. So $i^{50} = i^2 = -1$.
To find i^{101}: 101 ÷ 4 = 25 remainder 1. So $i^{101} = i^1 = i$.

Square roots of negative numbers:
$\sqrt{-9} = 3i$. $\sqrt{-8} = 2\sqrt{2}i$. $\sqrt{-1} = i$.

Example 2: Adding, Multiplying, and Dividing Complex Numbers

Let $z_1 = 2 + 3i$, $z_2 = 1 - 5i$.

Add: Pair the real parts, pair the imaginary parts.

$(2+1)+(3-5)i = 3-2i.$

Multiply: Distribute normally, then replace i^2 with -1.
$(2+3i)(1-5i) = 2-10i+3i-15i^2 = 2-7i+15 = 17-7i.$

Divide: Multiply top and bottom by the conjugate of the denominator.
$\frac{2+3i}{1-5i} \cdot \frac{1+5i}{1+5i} = \frac{2+10i+3i+15i^2}{1+25} = \frac{2+13i-15}{26} = \frac{-13+13i}{26} = -\frac{1}{2} + \frac{1}{2}i.$

Example 3: Conjugate and Modulus — The Twin Tools

For $z = a+bi$:

- **Conjugate**: $\bar{z} = a-bi$ (flip the sign of the imaginary part).
- **Modulus** (magnitude): $|z| = \sqrt{a^2+b^2}$ (distance from the origin).

For $z = 3+4i$: $\bar{z} = 3-4i$, $|z| = \sqrt{9+16} = 5$.

Key identity: $z \cdot \bar{z} = a^2+b^2 = |z|^2$.
Check: $(3+4i)(3-4i) = 9-16i^2 = 9+16 = 25 = 5^2$. Exactly.

Also: $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$. The conjugate of a product equals the product of the conjugates.

Example 4: The Complex Plane and Polar Form

Plot $a+bi$ as the point (a,b) in the plane. The distance from the origin is $r = |z|$. The angle measured from the positive x-axis is θ.

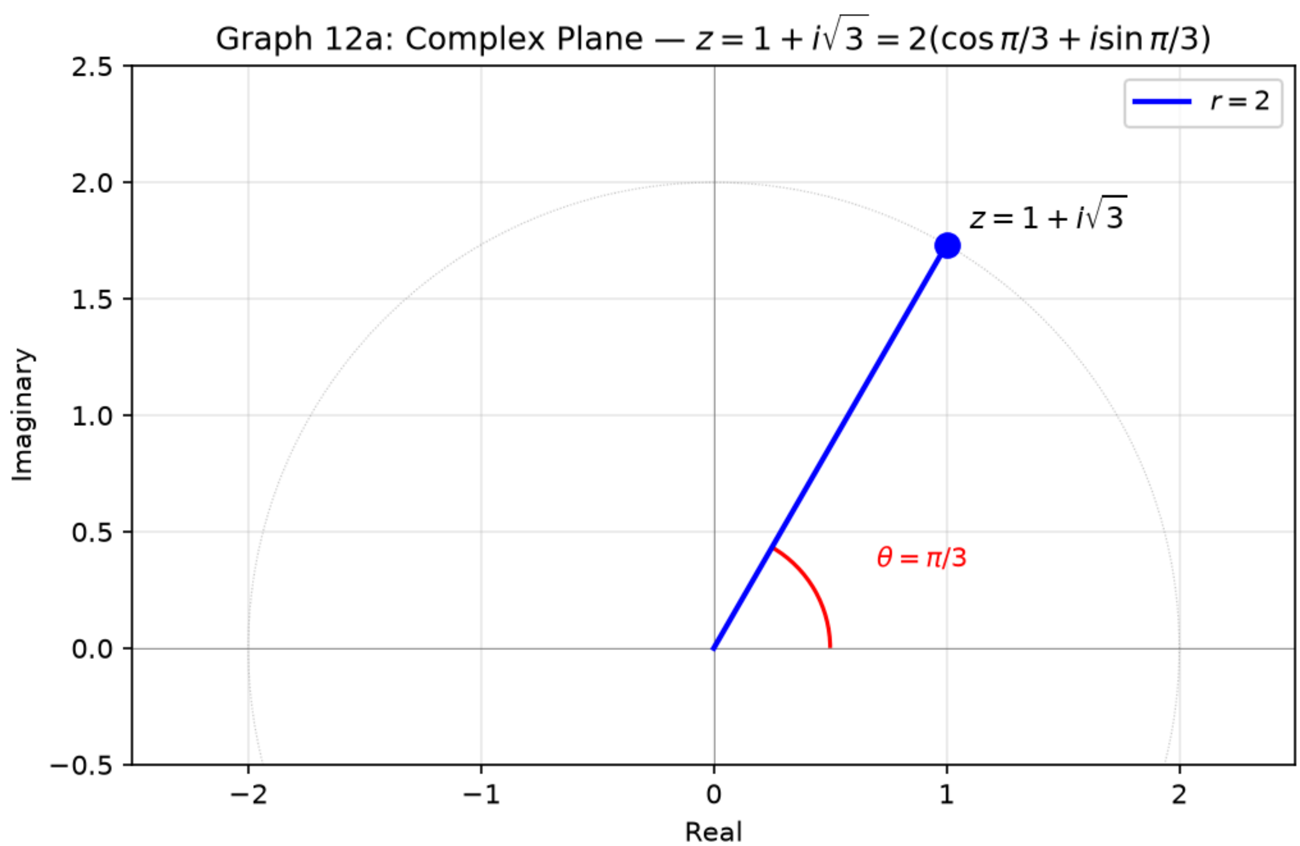
$a = r\cos\theta, b = r\sin\theta.$

This gives the **polar form**: $z = r(\cos\theta + i\sin\theta)$.

Example: $z = 1+i\sqrt{3}$.
$r = \sqrt{1+3} = 2$. $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, so $\theta = \frac{\pi}{3}$.

Polar form: $z = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$.

Example: $z = -1 + i$.
$r = \sqrt{2}$. $\cos\theta = -\frac{1}{\sqrt{2}}$, $\sin\theta = \frac{1}{\sqrt{2}}$, so $\theta = \frac{3\pi}{4}$.
Polar form: $z = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$.



Example 5: Euler's Formula and De Moivre — The Power Tools

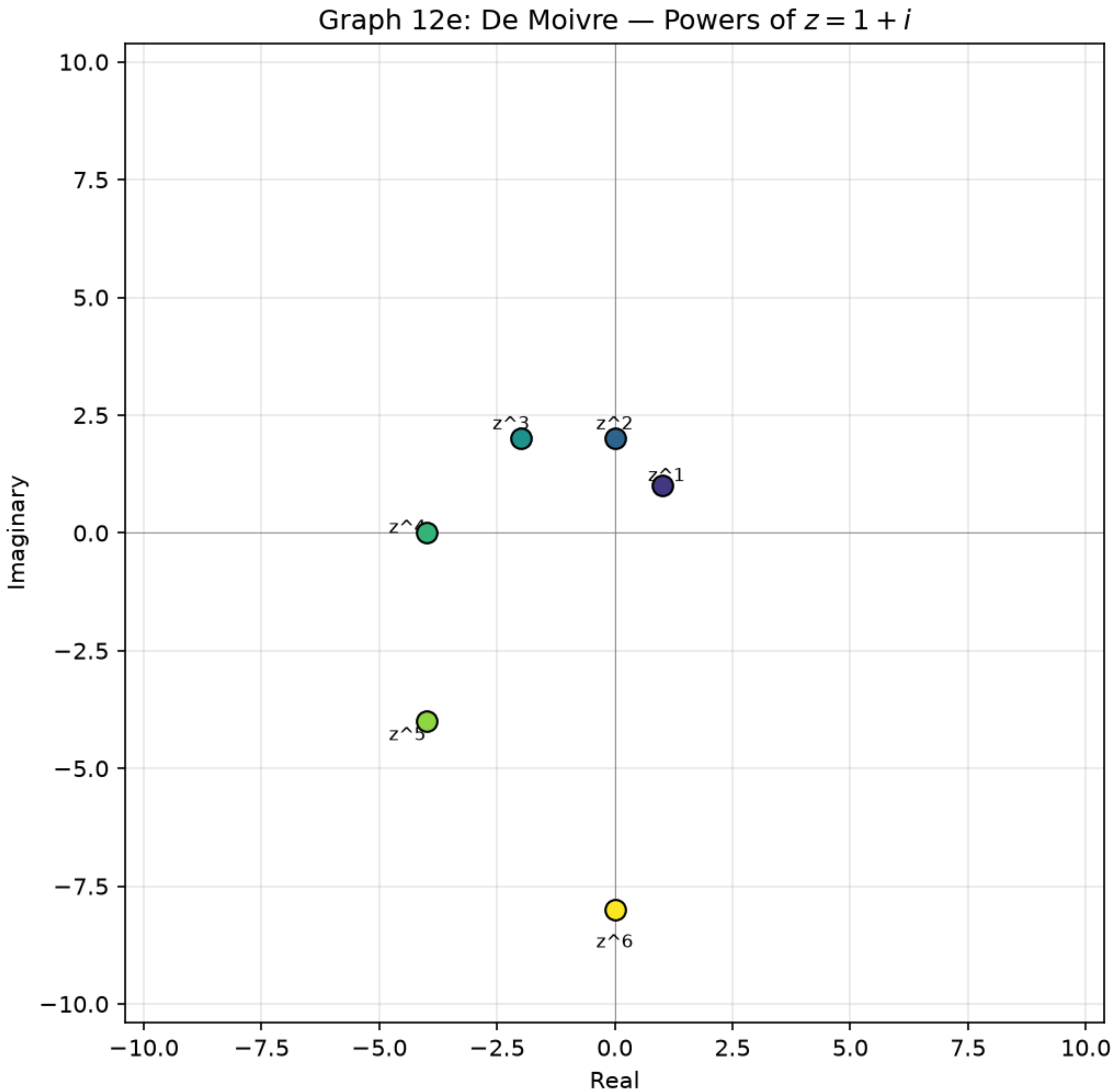
Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$.

This compresses the polar form to a single exponential: $z = re^{i\theta}$.

$1 + i\sqrt{3} = 2e^{i\pi/3}$. And the beautiful identity: $e^{i\pi} + 1 = 0$.

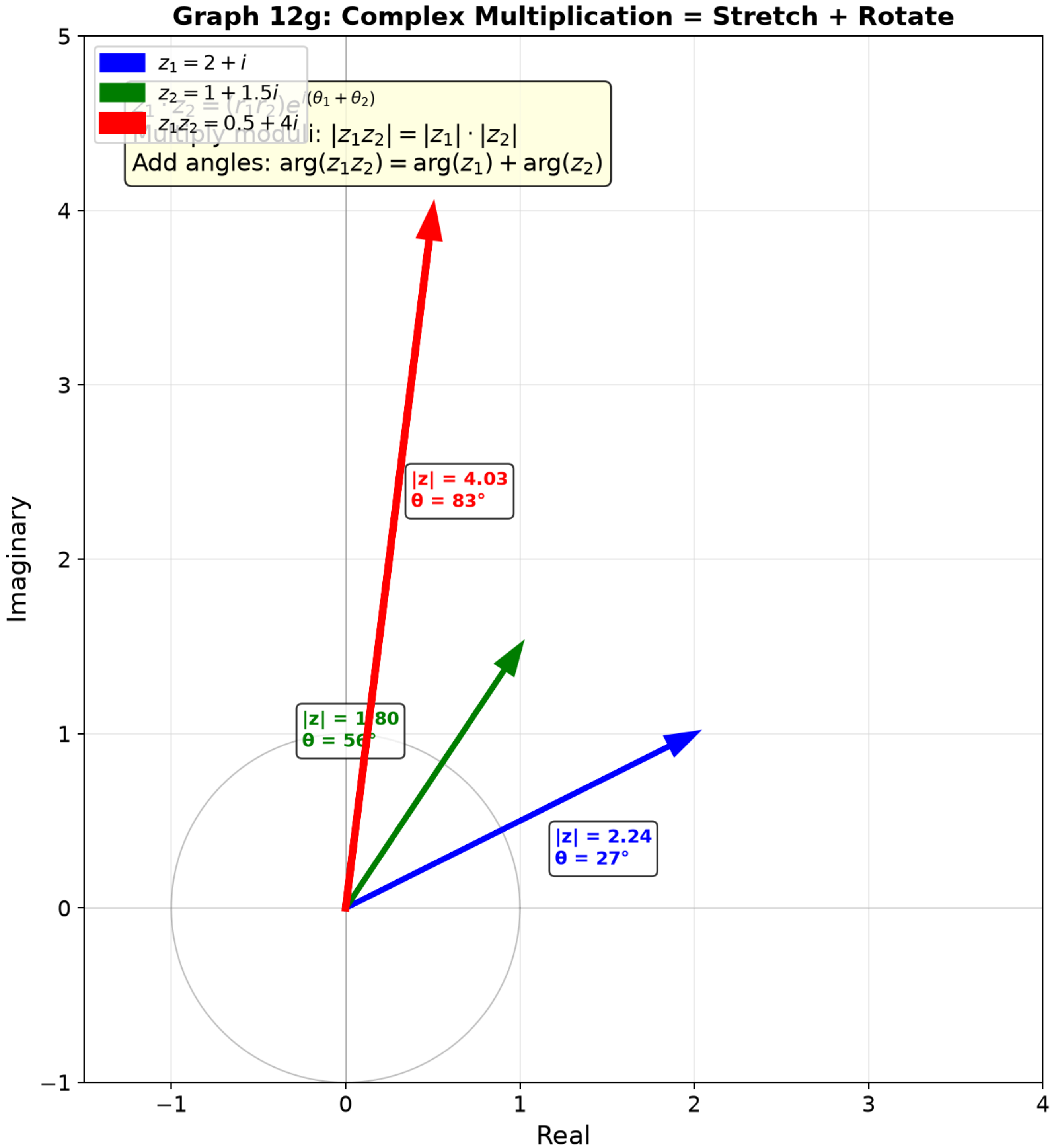
De Moivre's theorem: $z^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{in\theta}$.
Raising to a power multiplies the angle by n and raises the modulus by n.

Example: Compute $(1+i)^6$.
$r = \sqrt{1+1} = \sqrt{2}, \theta = \frac{\pi}{4}$.
$(1+i)^6 = (\sqrt{2})^6 \cdot e^{i \cdot 6\pi/4} = 8 \cdot e^{i \cdot 3\pi/2} = 8(0 - i) = -8i$.



Visual Insight — Complex Multiplication = Stretch + Rotate:

Every complex multiplication $z_1 \cdot z_2$ is a single geometric operation: stretch by $|z_2|$ and rotate by $\arg(z_2)$. This is why De Moivre works — raising to the nth power simply multiplies the angle by n and raises the modulus to the nth power.


Graph 12g: Complex Multiplication = Stretch + Rotate
$z_1 = 2 + i$
$z_2 = 1 + 1.5i$
$z_1 z_2 = 0.5 + 4i$
$z_1 \cdot z_2 = (r_1 r_2)e^{i(\theta_1+\theta_2)}$
Multiply moduli: $|z_1 z_2| = |z_1| \cdot |z_2|$
Add angles: $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
|z| = 4.03
θ = 83°
|z| = 1.80
θ = 56°
|z| = 2.24
θ = 27°
Imaginary
Real

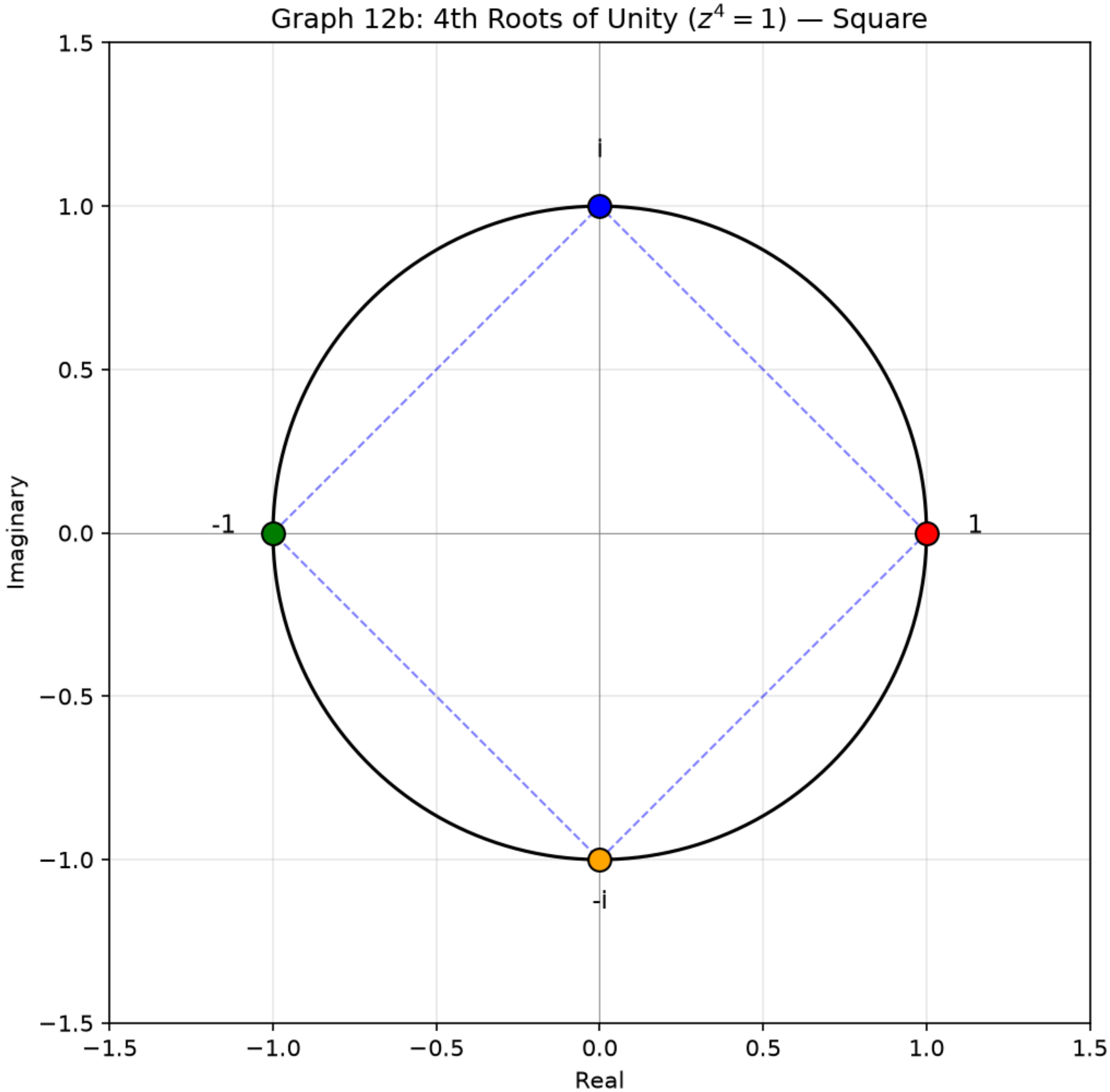
Example 6: nth Roots of Unity — Evenly Spaced Around the Circle

The equation $z^4 = 1$ has four solutions. Write $1 = e^{i \cdot 0}$:
$z_k = 1^{1/4} \cdot e^{i \cdot 2\pi k/4}$ for $k = 0, 1, 2, 3$.
→ $1,\ i,\ -1,\ -i$.

These four points form a square on the unit circle — evenly spaced by 90°.

Example: Solve $z^3 = -8$.
$-8 = 8e^{i\pi}$. The three cube roots:
$z_k = 2 \cdot e^{i(\pi + 2\pi k)/3}$ for $k = 0, 1, 2$.
$k = 0$: $2e^{i\pi/3} = 1 + i\sqrt{3}$.
$k = 1$: $2e^{i\pi} = -2$.
$k = 2$: $2e^{i \cdot 5\pi/3} = 1 - i\sqrt{3}$.

Three points equally spaced by 120° on a circle of radius 2.



> **Up to here**: $i^2 = -1$, 4-cycle. Conjugate flips the imaginary sign. Modulus = distance from origin.
> Polar form $re^{i\theta}$ packs both magnitude and direction. De Moivre handles powers in one step.
> The n nth roots are n points evenly spaced on a circle.

Visual Interlude: Three Geometric Views of Complex Numbers

View 1 — The Complex Plane as $\mathbb{R}^2$. A complex number $a + bi$ is exactly the point (a, b). Addition of complex numbers = addition of vectors. The modulus is the Euclidean distance from the origin.

View 2 — Polar Form as Stretch + Rotate. Multiplying by $re^{i\theta}$ stretches by factor r and rotates by angle θ. Complex multiplication is a single geometric operation combining scaling and rotation. This is why multiplying by i rotates by 90° — because $i = 1 \cdot e^{i\pi/2}$.

View 3 — Roots of Unity as Regular Polygons. The n solutions to $z^n = 1$ are n points equally spaced on the unit circle. They form a regular n-gon. Their sum is always 0 — the center of mass is at the origin.

Common Mistakes

Mistake 1: $\sqrt{-4} = -2$

Wrong path: "The square root of -4 is -2."

Why wrong: $(-2)^2 = 4$, not -4. The square root of a negative number is imaginary.

Right path: $\sqrt{-4} = \sqrt{4} \cdot \sqrt{-1} = 2i$. Always pull out the i.

Mistake 2: Forgetting the 4-cycle for powers of i

Wrong path: "$i^{23} = i^{20} \cdot i^3 = 1 \cdot i$... wait, what is i^3?"

Why wrong: Not internalizing the 4-cycle means recalculating every time.

Right path: Divide the exponent by 4. Remainder 0 → 1. Remainder 1 → i. Remainder 2 → -1. Remainder 3 → $-i$.

Mistake 3: Confusing $|z|^2$ with z^2

Wrong path: "$|3+4i|^2 = (3+4i)^2$."

Why wrong: $|z|^2 = z\bar{z} = a^2 + b^2 = 25$. But $z^2 = (3+4i)^2 = 9 + 24i - 16 = -7 + 24i$. Completely different.

Right path: $|z|^2$ is always a non-negative real number. z^2 is generally complex.

What We Just Did

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(1) Basic arithmetic — i² = −1 with 4-cycle. Add: pair real and imaginary parts.
    Multiply: distribute, replace i² → −1. Divide: multiply by conjugate of denominator.

(2) Geometric view — plot a+bi as (a,b). Modulus = distance from origin = |z|.
    Polar form z = re^{iθ} encodes magnitude r and direction θ.
    Multiplication: multiply moduli, add angles.

(3) Power tools — De Moivre: z^n = r^n e^{inθ}. Powers are one-step.
    nth roots: radius r^{1/n}, angles (θ+2πk)/n for k = 0,1,...,n−1.
    n roots form a regular n-gon on a circle.
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Practice 1

Divide $\frac{3-i}{2+i}$. Write the answer in the form $a + bi$.

→ Reference: **Example 2**

> Solutions: Solutions

Practice 2

Write $z = 1 - i$ in polar form. Then compute z^8 using De Moivre.

→ Reference: **Example 5**

> Solutions: Solutions

Practice 3

Find all three cube roots of -8. Give each in $a + bi$ form.

→ Reference: **Example 6**

> Solutions: Solutions

Practice 4: Composition

Explain why multiplying a complex number by i rotates it by 90° counterclockwise. Then find the result of multiplying $3 + 4i$ by i^3, and describe the geometric transformation in words.

→ Reference: **Example 1, 5**

> Solutions: Solutions

Practice 5

Write $z = -1 + i\sqrt{3}$ in polar form $re^{i\theta}$. Then find z^6.

→ Reference: **Example 4, 5**

> Solutions: Solutions

Practice 6: Real Battle

The three cube roots of -8 form a triangle in the complex plane. Find its area. Also find a 2×2 matrix that rotates the plane by $120°$, and verify that applying it three times gives the identity.

→ Reference: **Example 6, 13 (from 12A2)**

> Solutions: Solutions

Basic Algebra Drill — Complex Numbers (10 Problems)

> Pure calculation. Build fluency with i, conjugates, and polar form.

D1. Simplify i^{27}. Write as 1, -1, i, or $-i$.

D2. Compute $(2-3i)(1+4i)$. Write in $a+bi$ form.

D3. Find the conjugate and modulus of $z = 5 - 12i$.

D4. Find i^{4k+3} for any integer k. State the result.

D5. Simplify $\frac{1}{i}$. Write in $a+bi$ form.

D6. Compute $(3+2i)+(5-7i)-(1+4i)$.

D7. Find the modulus of $z = -6 + 8i$.

D8. Write $e^{i\pi/2}$ in $a+bi$ form.

D9. Find $\bar{z}$ and $|z|$ for $z = -3i$.

D10. Simplify $i^{15} + i^{16} + i^{17} + i^{18}$.

> Solutions: Solutions

Advanced Algebra Drill — Complex Numbers (10 Problems)

> Multi-step. Chain polar form, Euler, and De Moivre.

A1. Solve for z: $(1+i)z+3=2i-z$. Write z in $a+bi$ form.

A2. Compute $(1+i\sqrt{3})^9$ using De Moivre. Give the answer in $a+bi$ form.

A3. Find all four 4th roots of -16. Write each in polar form $re^{i\theta}$ and in $a+bi$ form.

A4. The four 4th roots of 1 form a square. Compute its area.

A5. Write $z=\sqrt{3}-i$ in polar form. Then compute z^6.

A6. Solve $z^2+2z+5=0$ over the complex numbers.

A7. Find all complex z such that $z^3=8i$.

A8. If $z=2e^{i\pi/6}$, find z^4 and $\frac{1}{z}$ in $a+bi$ form.

A9. Prove that $|z_1 z_2|=|z_1|\cdot|z_2|$ for any two complex numbers.

A10. The three cube roots of $8i$ form a triangle in the complex plane. Find its area.

> Solutions: Solutions

Today's Procedure

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Step 1: Arithmetic — i²=−1 with 4-cycle. Add: pair real, pair imaginary.
        Multiply: distribute, replace i²→−1. Divide: conjugate trick.
        Conjugate flips the sign of i. Modulus = distance from origin.

Step 2: Polar form — z = re^{iθ} where r = |z|, θ = arctan(b/a) with quadrant check.
        Multiplication: (r₁e^{iθ₁})(r₂e^{iθ₂}) = (r₁r₂)e^{i(θ₁+θ₂)}.
        De Moivre: (re^{iθ})^n = r^n e^{inθ}.

Step 3: Roots — z^n = w has exactly n solutions.
        z_k = |w|^{1/n} · e^{i(arg(w) + 2πk)/n}, k = 0,1,...,n−1.
        The n roots are evenly spaced on a circle — a regular n-gon.
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How to Read These Symbols

Symbol	Reads as	Meaning
i	"i" / "the imaginary unit"	$i^2 = -1$ — fundamental imaginary number
$z = a + bi$	"z equals a plus b i"	complex number: a=real part, b=imaginary part
$\bar{z}$	"z bar" / "z conjugate" / "complex conjugate"	$\overline{a+bi} = a - bi$ — flip sign of imaginary part
$	z	$
$\mathrm{Re}(z)$	"real part of z"	the a in a+bi
$\mathrm{Im}(z)$	"imaginary part of z"	the b in a+bi (a real number — NOT bi!)
$re^{i\theta}$	"r e to the i theta" / "polar form"	$r=

Symbol	Reads as	Meaning
$\arg(z)$	"argument of z"	angle θ from positive real axis
Euler's formula	"Euler's formula"	$e^{i\theta} = \cos\theta + i\sin\theta$
De Moivre	"De Moivre's theorem"	$z^n = r^n e^{in\theta} = r^n(\cos n\theta + i\sin n\theta)$
n-th roots of unity	"n-th roots of unity"	$e^{2\pi ik/n}$ for k=0,1,...,n-1 — equally spaced on unit circle
complex plane / Argand diagram	"complex plane" / "Argand diagram"	x-axis=real, y-axis=imaginary

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Terminology

Up to now we used plain words like "imaginary", "conjugate", "modulus", "angle".
You have already learned all the methods. Now we attach the formal mathematical names.

What we called it	Mathematical term	Notation
imaginary unit	imaginary unit	$i, i^2 = -1$
conjugate	complex conjugate	$\bar{z} = a - bi$
modulus / magnitude	modulus	$
argument / angle	argument	$\theta = \arg(z)$
polar form	polar / exponential form	$z = re^{i\theta}$
Euler's formula	Euler's formula	$e^{i\theta} = \cos\theta + i\sin\theta$
De Moivre's theorem	De Moivre's theorem	$z^n = r^n e^{in\theta}$
nth roots of unity	roots of unity	$e^{2\pi ik/n}, k = 0, \dots, n-1$